

Goal-Oriented Autonomous Robotic Car with Obstacles Avoidance and Box Retrieval System

Md.Sajjad Hossain Mridul

*Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2006036@eee.buet.ac.bd*

S.M. Mahfuz

*Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2006060@eee.buet.ac.bd*

Emam Al Mahdee

*Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2006048@eee.buet.ac.bd*

Md. Shakib Hossain

*Department of Electrical and Electronic Engineering
Bangladesh University of Engineering and Technology
Dhaka, Bangladesh
2006065@eee.buet.ac.bd*

Abstract—This project will explore the real-time path planning algorithm in the well-known environment using an ultrasonic sensor and use of a push-box mechanism to perform certain tasks. We will test our algorithm on a small two-dimension grid-based environment for the limitation of computation ability and storage capability for the microcontroller. Also to get feedback from the grid, we are going to use an IR sensor to sense the black color of every node and update the current position, only turning 90 degrees left or right, and moving forward, backward, right, and left on the grid based environment and a push box mechanism that can push certain objects to the robotic car. This project aims to gain the experience and ability to implement an academic algorithm in real-world products.

I. INTRODUCTION

In the realm of robotics, real-time path planning and obstacle avoidance have become critical challenges in creating effective autonomous systems. Our project focuses on building an autonomous robotic car equipped with an ultrasonic sensor for real-time navigation and a push box mechanism for object retrieval tasks in a controlled, two-dimensional, grid-based environment. This project is a blend of theoretical algorithm development and practical implementation, aimed at bringing academic innovations into real-world applications. [1], [2]

The primary motivation behind this endeavor is to develop a reliable and efficient system capable of maneuvering autonomously, making split-second decisions to navigate obstacles, and accomplishing defined tasks with minimal human intervention. By leveraging the capabilities of ultrasonic sensors, our robotic car will detect and avoid obstacles while adhering to a predetermined path-planning algorithm, enhancing the system's overall adaptability and precision.

Additionally, to optimize computational efficiency and storage limitations imposed by microcontroller hardware, we have restricted movements to 90-degree turns and basic directional steps (forward, backward, left, and right) on the grid. This structured approach reduces unnecessary computations, mak-

ing the algorithm more scalable and adaptable within resource constraints.

Furthermore, integrating a push mechanism with the autonomous car extends the system's capabilities beyond navigation. The push mechanism's ability to push the object into specified car adds a layer of versatility to the system, simulating scenarios commonly faced in logistics, automation, and industrial applications.

II. METHODOLOGY

A. Block Diagram and Brief Description

The project involves a 6x6 grid system where each node has two identifiers: a number from 0 to 35 for positioning, and a state indicator (0 for empty, 1 for optimal, 2 for obstacle, 3 for goal). The A* search algorithm is used to calculate an optimal path from a starting node to a goal node by minimizing movement costs (f-cost), which is the sum of g-cost (distance from start) and h-cost (estimated distance to goal). [3]The robotic car navigates this path, avoiding obstacles using an ultrasonic sensor and getting feedback from IR sensor. If a new obstacle is detected, the path is recalculated using A*. Once at the goal, the car gets an item and returns to the start by retracing its path. [4]

B. Description of The Block Diagram

We will take user input using bluetooth module for the goal position and feed it to Arduino. Arduino will execute A* search algorithm to find the optimum path. It will serial print this. Once optimal path list is found it will follow that path list until reaching the goal position. It will update its current position using the feedback from IR sensor. If it detects any obstacle on its way, it will recalculate the path list.

After reaching the goal position, it will trigger the push box mechanism. The push mechanism contains another ultrasonic sensor to sense the car reached. If it senses the robotic car for 5 seconds, it will trigger the servo motor, the servo motor

35	34	33	32	31	30
29	28	27	26	25	24
23	22	21	20	19	18
17	16	15	14	13	12
11	10	9	8	7	6
5	4	3	2	1	0

Fig. 1. Proposed grid framework

will extend, and objects will fall into the car. The car is surrounded by a safety rail. This design ensures that the object remains safely enclosed during transport, mitigating any risk of displacement or loss during the car's subsequent movements.

Once retrieved, the car will follow the return path list and reaching source node.

Implementing the push box mechanism for our autonomous robotic car presents several key challenges that must be addressed to ensure optimal performance and reliability.

The first challenge lies in accurately reaching the target place. Navigating the robotic car to position precisely in front of the object requires meticulous path planning and sensor accuracy, particularly in a dynamic environment where obstacles may impede movement.

The second challenge is the effective pushing of the object using the servo motor. Successful alignment and securing of the object requires precise control of the servo motor. Any misalignment or instability during the pushing process can lead to task failure.

3	0	0	0	2	0
2	0	2	0	0	0
0	0	0	0	0	2
0	2	0	2	0	0
0	0	0	0	0	0
0	0	2	0	0	1

Fig. 2. Predefined representation of obstacles

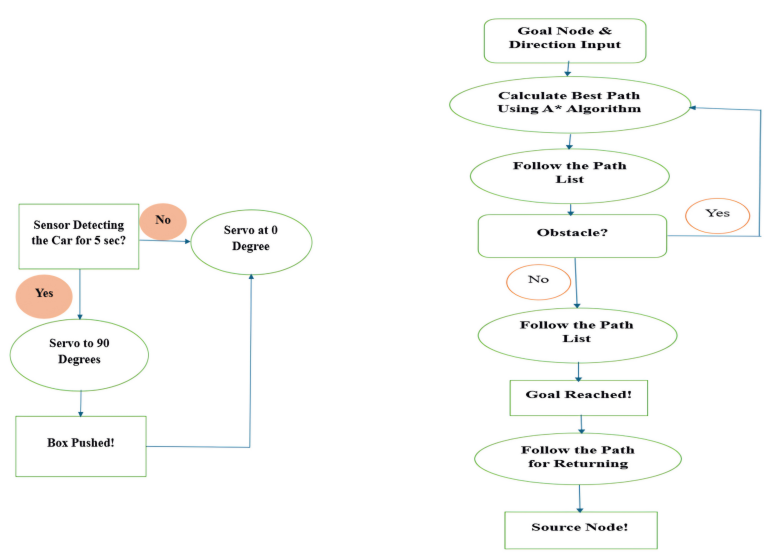


Fig. 3. Algorithm of proposed model

Lastly, the shape of the object presents an inherent challenge. The push mechanism has been designed with specific dimensions and geometries in mind, making it less effective at securing objects with irregular shapes or dimensions that fall outside of its optimized range. This constraint necessitates careful consideration of target objects during both design and operational phases.

III. COST BUDGET

IV. RESULT

When we start the car at every step mechanical error is added to our program. So, in the final step we must re-organize our car to retrieve the object. Real time object avoidance works fine in our project.

V. CONCLUSION

In this project, we achieved a working real-time path-planning system for a robotic car equipped with an ultrasonic sensor and a push box mechanism. The car successfully navigated a controlled, grid-based environment by moving in

TABLE I
COST BUDGETING

Components	Quantity	Estimated Cost(Taka)
Robot car chassis	1	1000
Arduino Mega	2	2800
Motor Driver	3	1275
DC Gear Motor	4	3100
Ultrasonic sensor	1	75
Batteries and Holder	1	4000
Servo Motor	2	800
wires	-	200
Linea Actuator	1	4000
Total		18050

feedback based grid system, which simplified pathfinding and reduced the need for intensive computation. This grid-based layout allowed the robot to detect obstacles, plan a safe path, and retrieve specific objects—showcasing how path-planning algorithms can be applied in practical situations.

By using an ultrasonic sensor for obstacle detection, we effectively avoided collisions, allowing the robot to respond in real-time. While the project environment was simple, our approach proved effective, demonstrating how such a system could potentially be adapted for larger, more complex scenarios. However, some limitations remain: the system could benefit from more advanced algorithms to handle dynamic environments and overcome potential obstacles more smoothly.

Through this work, we gained valuable hands-on experience with implementing academic theories in a real-world project. These results show how real-time path-planning, combined with object-handling mechanisms, can be practically applied, offering potential for future improvements and applications. In future projects, incorporating machine learning or adaptive algorithms could make the robot more responsive, allowing it to navigate complex spaces and handle varied tasks with greater efficiency. Also, for now, we are planning the path in a pre-installed way. In future, we can use the robot car to map the whole area for obstacles and then upload the map in its memory. Thus, this project can be made more dynamic and self-sufficient. Also, our program works to minimize the path. We can program it to use the shortest time approach to reach the destination and save valuable time in the industry.

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